

COMS20006

SOFTWARE ENGINEERING PROJECT

AWS Overview

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What is AWS?

AWS is a cloud computing platform that provides a wide range of cloud-based services and resources for building, deploying, and managing applications and infrastructure.

You might have some experience of AWS from CSA.



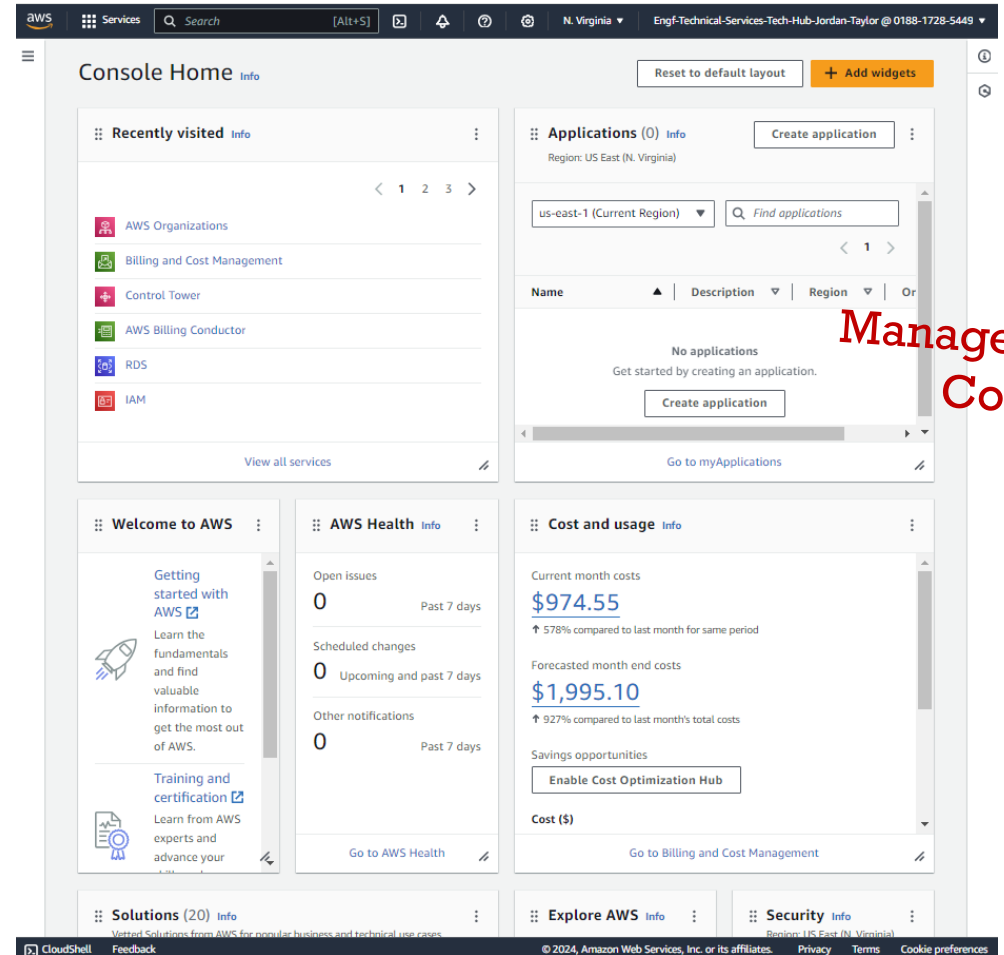
What is AWS?

As a cloud developer in SEP, you are probably going to be most interested in these two features.

- AWS offers three primary service models:
- **Infrastructure as a Service (IaaS):** Virtual machines, storage, and networking resources are available for you to manage and configure as needed. It's like having a virtual data center in the cloud.
- **Platform as a Service (PaaS):** AWS provides pre-configured platforms for application development, reducing the need to manage underlying infrastructure. Examples include AWS Amplify for web applications and AWS Lambda for serverless computing.
- **Software as a Service (SaaS):** AWS does not specialize in offering SaaS. The most well-known example would be Microsoft's Azure. Azure hosts and manages software applications that users can access over the internet, such as Office 365.

NOTE:

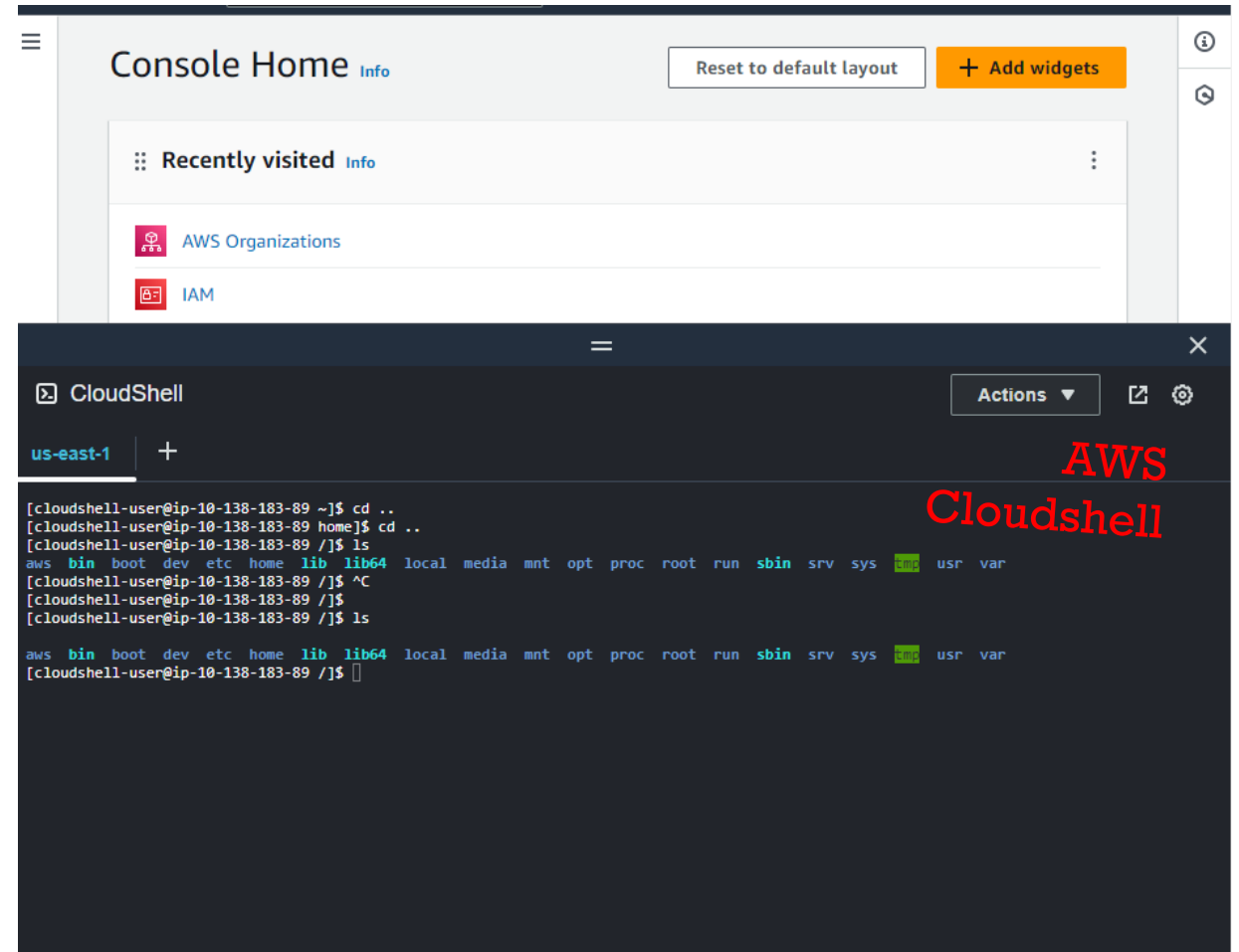
You can make use of AWS services through the **Management Console**, command line interface (CLI), or programmatically via **SDK**.



Management
Console

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<https://boto3.amazonaws.com/v1/documentation/api/latest/index.html>

The screenshot shows the official AWS SDK for Python (Boto3) website. The browser address bar displays `aws.amazon.com/sdk-for-python/`. The AWS navigation bar is at the top, featuring the AWS logo, links for 'About AWS', 'Contact Us', 'Support', 'English', and 'My Account', along with a 'Sign In to the Console' button. Below the navigation bar, a sidebar on the left contains 'RESOURCES' (with 'AWS SDK for Python' selected), 'RELATED LINKS' (with 'Documentation' and 'Tools'), and a 'Get started for free' section with a 'Create free account' button. The main content area is titled 'AWS SDK for Python (Boto3)' and includes a brief introduction, an 'Install' section with the command `pip install boto3`, and links for 'Getting Started', 'API Reference', and 'Community Forum'. A 'Key features' section is also visible at the bottom, mentioning 'Resource APIs'.

Official SDK
(python boto3)

AWS SDK for Python (Boto3)

Get started quickly using AWS with `boto3`, the AWS SDK for Python. Boto3 makes it easy to integrate your Python application, library, or script with AWS services including Amazon S3, Amazon EC2, Amazon DynamoDB, and more.

Install

```
pip install boto3
```

Or get the latest tarball on PyPI

Find the source on GitHub »

Key features

Resource APIs

Boto3 has two distinct levels of APIs. Client (or "low-level") APIs provide one-to-one mappings to the underlying HTTP API operations. **Resource APIs** hide explicit network calls but instead provide resource objects and collections to access attributes and perform actions. For example:

NOTE:

You can make use of AWS services through the **Management Console**, **command line interface (CLI)**, or programmatically via **SDK**.

Note that multiple SDKs are available for different languages/frameworks.

<https://aws.amazon.com/sdk-for-net/>

Tool to Build on AWS

AWS SDK for .NET

Develop and deploy applications with the AWS SDK for .NET. The SDK makes it easy to call AWS services using idiomatic .NET APIs.

Get started with SDK for .NET

Install from NuGet

Official SDK
(.NET)

How it works

AWS SDK for .NET simplifies the use of AWS services by providing a set of libraries that are consistent and familiar for .NET developers. All AWS SDKs provide support for API lifecycle consideration such credential management, retries, data marshaling, and serialization. SDK for .NET also supports for higher level abstractions such as the Amazon S3 Transfer Utility, Amazon Cognito Identity Provider, and AWS DynamoDB Session State provider. Visit [aws/dotnet Github](https://aws.amazon.com/dotnet) for .NET tools and libraries on AWS.



Section II

“As a Service”

	Traditional IT	IaaS	PaaS	SaaS
Applications	You manage	You manage	You manage	Provider manages
Data	You manage	You manage	You manage	Provider manages
Runtime	You manage	You manage	Provider manages	Provider manages
Middleware	You manage	You manage	Provider manages	Provider manages
OS	You manage	Provider manages	Provider manages	Provider manages
Virtualization	You manage	Provider manages	Provider manages	Provider manages
Servers	You manage	Provider manages	Provider manages	Provider manages
Storage	You manage	Provider manages	Provider manages	Provider manages
Networking	You manage	Provider manages	Provider manages	Provider manages

 You manage  Provider manages

Infrastructure as a Service

With IaaS, you can configure and control the infrastructure, install and manage software, and have full control over your resources as if you were managing physical hardware, but without the need to maintain the hardware.

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 You manage  Provider manages

Platform as a Service

With PaaS, tasks such as operating system management, scalability, and security are managed for you.

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 You manage  Provider manages

Software as a Service

With SaaS, fully fledged software is delivered to the end-user/developer, usually through a browser.

You probably use SaaS every day, with software such as Office 365.

	Traditional IT	IaaS	PaaS	SaaS
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Infrastructure as a Service - VMs

- Virtual Machines ←
- Storage
- Virtual networks

Virtual Machines:

Described as ‘*EC2 instances*’ in **AWS**.

Can be provisioned with a variety of operating systems.

Come in a variety of sizes and costs.

EC2 > Instances > Launch an instance

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

 [Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Quick Start

Amazon Linux
aws

macOS
Mac

Ubuntu
ubuntu®

Windows
Microsoft

Red Hat
Red Hat

SUSE L
SUSE


Browse more AMIs
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI
ami-0b72821e2f351e396 (64-bit (x86), uefi-preferred) / ami-07fa59ce6da2fe3fe (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible ▼

Description

Amazon Linux 2023 is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Architecture	Boot mode	AMI ID	
64-bit (x86) ▼	uefi-preferred	ami-0b72821e2f351e396	Verified provider

Infrastructure as a Service - VMs

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OS List

	Amazon Linux 2023 AMI ami-0b72821e2f351e396 (64-bit (x86), uefi-preferred) / ami-07fa59ce6da2fe3fe (64-bit (Arm), uefi) Amazon Linux 2023 is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications. Platform: amazon Root device type: ebs Virtualization: hvm ENA enabled: Yes
	Amazon Linux 2 AMI (HVM) - Kernel 5.10, SSD Volume Type ami-01fccab91b456acc2 (64-bit (x86)) / ami-08293cd37d25b872d (64-bit (Arm)) Amazon Linux 2 comes with five years support. It provides Linux kernel 5.10 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras. This AMI is the successor of the Amazon Linux AMI that is now under maintenance only mode and has been removed from this wizard. Platform: amazon Root device type: ebs Virtualization: hvm ENA enabled: Yes
	macOS Sonoma ami-0c4b66d9dadb7e395 (64-bit (Mac)) / ami-0c45f2bf394dee00c (64-bit (Mac-Arm)) The macOS Sonoma AMI is an EBS-backed, AWS-supported image. This AMI includes the AWS Command Line Interface, Command Line Tools for Xcode, Amazon SSM Agent, and Homebrew. The AWS Homebrew Tap includes the latest versions of multiple AWS packages included in the AMI. Platform: x86_64_mac Root device type: ebs Virtualization: hvm ENA enabled: Yes
	macOS Ventura ami-0c5ac9ce557a3477 (64-bit (Mac)) / ami-01aa3973cdaf40134 (64-bit (Mac-Arm)) The macOS Ventura AMI is an EBS-backed, AWS-supported image. This AMI includes the AWS Command Line Interface, Command Line Tools for Xcode, Amazon SSM Agent, and Homebrew. The AWS Homebrew Tap includes the latest versions of multiple AWS packages included in the AMI. Platform: x86_64_mac Root device type: ebs Virtualization: hvm ENA enabled: Yes
	Red Hat Enterprise Linux 9 (HVM), SSD Volume Type ami-0585d8c7a9c35822c (64-bit (x86)) / ami-07472131ec292b5da (64-bit (Arm)) Red Hat Enterprise Linux version 9 (HVM), EBS General Purpose (SSD) Volume Type Platform: rhel Root device type: ebs Virtualization: hvm ENA enabled: Yes
	macOS Monterey ami-069e306e29db9ed55 (64-bit (Mac)) / ami-03539bbaad2b27b75 (64-bit (Mac-Arm)) The macOS Monterey AMI is an EBS-backed, AWS-supported image. This AMI includes the AWS Command Line Interface, Command Line Tools for Xcode, Amazon SSM Agent, and Homebrew. The AWS Homebrew Tap includes the latest versions of multiple AWS packages included in the AMI. Platform: x86_64_mac Root device type: ebs Virtualization: hvm ENA enabled: Yes
	Red Hat Enterprise Linux 9 with High Availability ami-036c2987def867fb (64-bit (x86)) Red Hat Enterprise Linux version 9 with High Availability (HVM), EBS General Purpose (SSD) Volume Type

Infrastructure as a Service - VMs

VM Size List

- Virtual Machines
- Storage
- Virtual networks

Virtual Machines:

Described as 'EC2 instances'

Can be provisioned with various operating systems.

Come in a variety of sizes

Viewing 726 of 726 available instances <- Many choices!

< 1 2 3 4 5 6 7 ... 37 >

Instance name ▲	On-Demand hourly rate ▼	vCPU ▼	Memory ▼	Storage ▼	Network performance ▼
t4g.nano	\$0.0042	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t4g.micro	\$0.0084	2	1 GiB	EBS Only	Up to 5 Gigabit
t4g.small	\$0.0168	2	2 GiB	EBS Only	Up to 5 Gigabit
t4g.medium	\$0.0336	2	4 GiB	EBS Only	Up to 5 Gigabit
t4g.large	\$0.0672	2	8 GiB	EBS Only	Up to 5 Gigabit
t4g.xlarge	\$0.1344	4	16 GiB	EBS Only	Up to 5 Gigabit
t4g.2xlarge	\$0.2688	8	32 GiB	EBS Only	Up to 5 Gigabit
t3.nano	\$0.0052	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3.micro	\$0.0104	2	1 GiB	EBS Only	Up to 5 Gigabit
t3.small	\$0.0208	2	2 GiB	EBS Only	Up to 5 Gigabit

Cheapest instances

Infrastructure as a Service - VMs

VM Size List

- Virtual Machines
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Instance name ▾	On-Demand hourly rate ▾	vCPU ▾	Memory ▾	Storage ▾	Network performance ▾
u-12tb1.112xlarge	\$109.20	448	12288 GiB	EBS Only	100 Gigabit
p5.48xlarge	\$98.32	192	2048 GiB	8 x 3840 GB SSD	3200 Gigabit
u-9tb1.112xlarge	\$81.90	448	9216 GiB	EBS Only	100 Gigabit
u-6tb1.112xlarge	\$54.60	448	6144 GiB	EBS Only	100 Gigabit
u-6tb1.56xlarge	\$46.40391	224	6144 GiB	EBS Only	100 Gigabit
p4d.24xlarge	\$32.7726	96	1152 GiB	8 x 1000 SSD	400 Gigabit
u-3tb1.56xlarge	\$27.30	224	3072 GiB	EBS Only	50 Gigabit
x1e.32xlarge	\$26.688	128	3904 GiB	2 x 1920 SSD	25 Gigabit
x2iedn.32xlarge	\$26.676	128	4096 GiB	2 x 1900 NVMe SSD	100 Gigabit
x2iedn.metal	\$26.676	128	4096 GiB	2 x 1900 NVMe SSD	100 Gigabit
trn1n.32xlarge	\$24.78	128	512 GiB	4 x 1900 NVMe SSD	1600 Gigabit
p3.16xlarge	\$24.48	64	488 GiB	EBS Only	25 Gigabit

Be careful!
Things can get very expensive.

Virtual Machines:

Described as 'EC2 instances'

Can be provisioned with a variety of operating systems.

Come in a variety of sizes

**MOST
EXPENSIVE
instances**

Infrastructure as a Service - Storage

- Virtual Machines
- Storage
- Virtual networks



Infrastructure as a Service - Storage

- Virtual Machines
- Storage
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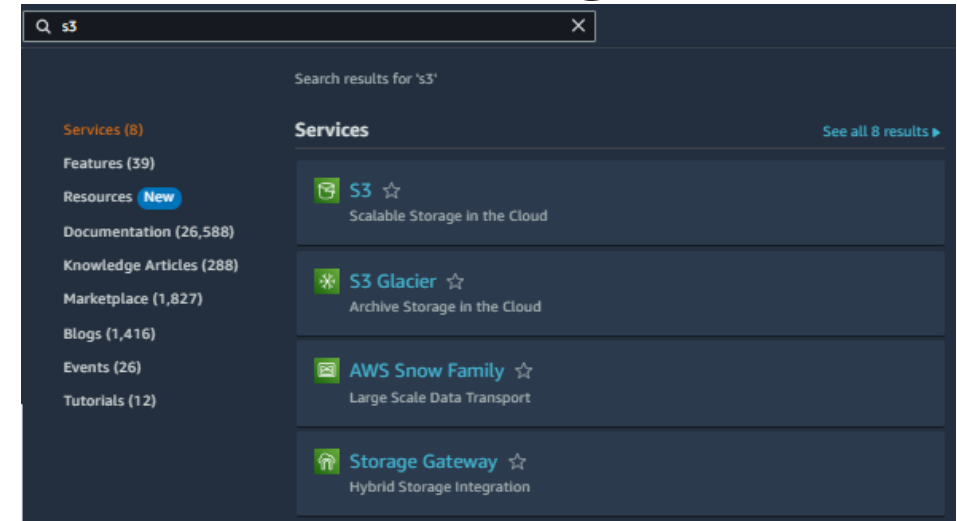
S3 (Storage):

Stores data and files.

Note that this is **not** a database.

Data objects receive a unique namespace, accessible anywhere using HTTP/HTTPS.

Accessible via REST API – wrapped in AWS SDK (boto3).



Create a bucket

Every object in S3 is stored in a bucket. To upload files and folders to S3, you'll need to create a bucket where the objects will be stored.

Create bucket

Infrastructure as a Service - Storage

- Virtual Machines
- Storage
- Virtual networks



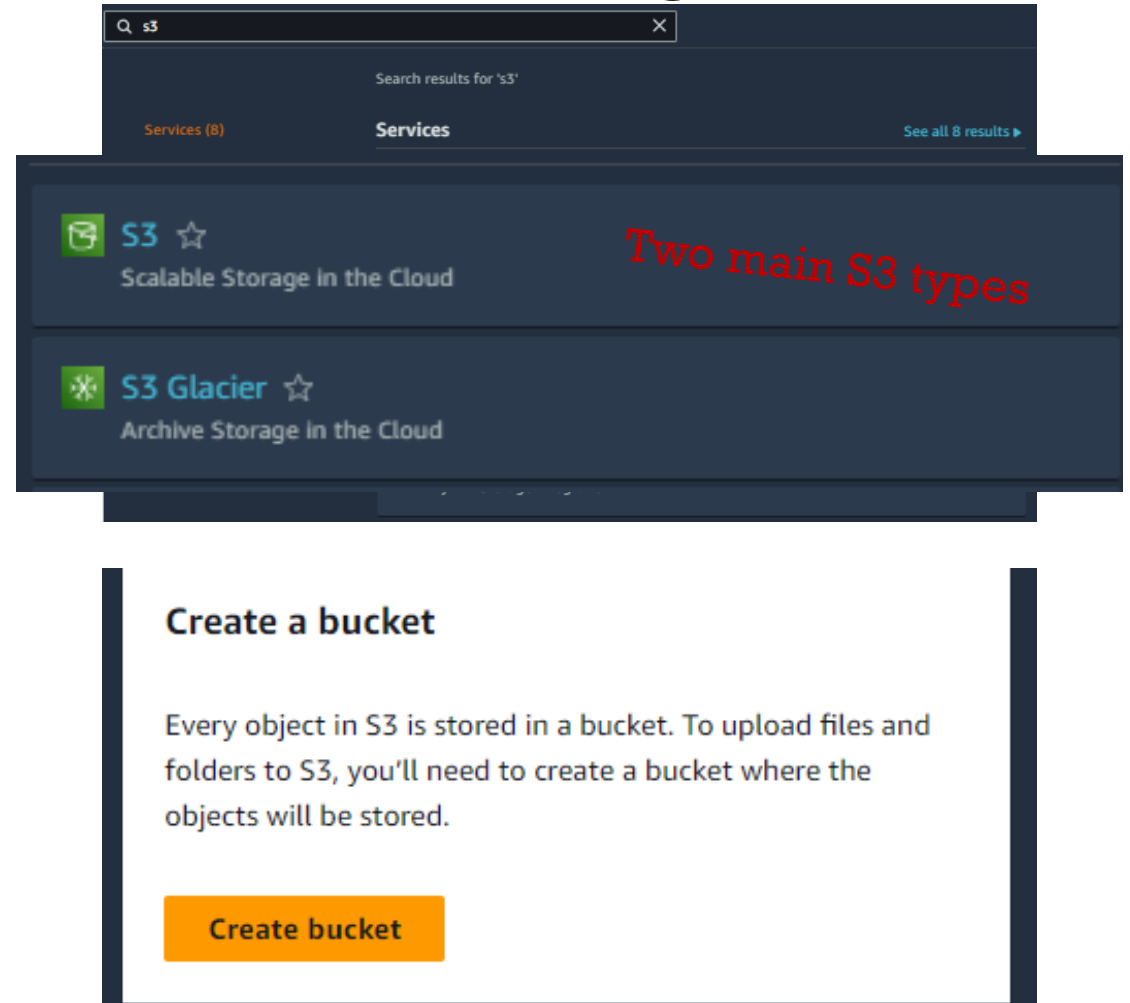
S3 (Storage):

Stores data and files.

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Data objects receive a unique namespace, accessible anywhere using HTTP/HTTPS.

Accessible via REST API – wrapped in AWS SDK (boto3).



Infrastructure as a Service – Storage (Databases)

- Virtual Machines
- Storage
- Virtual networks



DynamoDB:

Serverless, NoSQL, fully managed database.

Non-relational (flat).

Quick setup

Extremely scaled – elastic throughput.



Amazon
DynamoDB

Infrastructure as a Service – Storage (Databases)

- Virtual Machines
- Storage
- Virtual networks



Amazon RDS (Relational Database Service):

Managed SQL Database Service.

Relational.

Choice of engine (Aurora, MySQL, Oracle, PostgreSQL, MariaDB...)

Can be backed up to S3.

Scaled – elastic throughput.



**Amazon
RDS**

Infrastructure as a Service – Storage (Queues)

- Virtual Machines
- Storage
- Virtual networks



Queue Storage:

All-in-one message broker and storage service.

Accessible start and end points across the world, using calls over HTTP or HTTPS.

A queue message can be up to 256 KiB in size.

Useful to create a backlog of work to process asynchronously.

Larger latency than similar technologies such as RabbitMQ, but managed scaling.

Infrastructure as a Service - Virtual Networks

- Virtual Machines
- Storage Accounts
- Virtual networks



Virtual Networks:

Essentially a virtual LAN between your virtual machines.

Allows secure communication between VMs.

*Probably not
required in SEP*

Platform as a Service + Serverless

AWS Lambda (Serverless)

Note: There exists many more services, but they exist outside the scope of this lecture / are not applicable to most groups.
The line between IaaS and PaaS can be *grey*.

E.g. Kubernetes Service Clusters

Platform as a Service + Serverless

AWS Lambda (Serverless)



You write the code, define the trigger, and AWS handles the rest.

Convenient method to build and deploy microservices, automation, and event-driven applications in the cloud

AWS Lambda:

Serverless, event-driven functions that can execute code in response to various triggers without requiring you to manage the infrastructure.

Implicitly scales based on demand.

